

SCI-MAP

Ordinary Mapmaking and Observation Coaddition

Formal Scientific/Engineering Contract

Scientific contract version: **v0.2**
 Document revision: **r0.1**
 Status: **Scope-approved successor candidate; scientific-owner disposition pending**
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Authority boundary. This is the normative formal view of the implementation-independent scientific contract. It preserves the canonical equations, 52 requirements, 25 predictions, exact state and support semantics, provenance, and conformance routing. The separate science-team rationale r0.1 explains the same science without replacing this authority. This document does not claim implementation conformity, representation fidelity, scientific validation, observational performance, or production readiness.

Physical model and contract at a glance

MAP consumes the realized MAP-facing output of PTC; it never bypasses PTC to consume CAL output directly. ALIGN/AST supplies the coordinate facts and MAP supplies the target grid and declared projection. SCI-MAP applies one fixed, positive-coefficient linear averaging operator to the signal and to every declared linear companion. A coadd then applies a second positive-coefficient operator to atomically admitted, compatible observation bundles on the same grid. All scientific statements are conditional on the complete admitted calibration, alignment, eligibility, coefficient, response, geometry, and covariance state.

Facet	Compact authority
Input	Realized SCI-PTC v0.1/r0.5 calibrated- x -derived nonpolarimetric total-intensity-equivalent product in mJy/beam , with exact originating nominal-beam identity, occurrence/generation, retention, causes, and response state; exact AST RTC-grid coordinates for the same stable sample and parent; and an explicitly selected, completely handed-off admitted PTC MAP coefficient family.
Output	A complete immutable observation-map bundle and, when requested and admissible, a complete centered-integer common-grid observation coadd bundle.
Equation	On admitted membership, $N_p = \sum_i a_{pi} z_i$ and $Q_p = \sum_i a_{pi}$. The normalized scientific vector contains exactly the rows authorized by the effective support policy, with $\hat{m}_p = N_p/Q_p$ there; response and covariance use the same explicit row selection. Coadd repeats this restricted structure over complete observations.
Source	Frozen SCI-PTC v0.1/r0.5 digest 8357961a49272adc40e27a8aa9e760e0d01ff2419ae2c88a62c0f93c9f959e66; SCI-CAL v0.1 science r0.5/engineering r0.4 digest 413426f49edf1249f751a05bb8c6e9fd907b11e8da0530fe2da39814885efb22; SCI-AST v0.1/r0.3 manifest digest b54b6013750540f28aad02339a60bf36078980dc53b132beab73069d66ef3601; SCI-VAL Core v0.1/r0.3 digest 2fc3b3ad329fe3035d442b43d1e564a74fc86ab49f85f56e87322d8553fad9a6; SCI-PTC_TO_SCI-MAP_BOUNDARY v0.2-draft.1/r0.4; the admitted uniform Registry/family/profile and uniform VAL binding; and the exact inherited MAP/AST profiles and boundaries. The final v0.2 source/VAL binding generation remains pending.
Status	Contract author draft. Algebraic predictions are preregistered below; none has been executed here. All stronger claim layers remain unassessed.

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1 Scientific boundary and estimand

The ordinary observation map estimates the coefficient-weighted average of the admitted realized PTC sample quantity in each declared map pixel. Its meaning is inherited through the required CAL-to-PTC route, from ALIGN/AST's coordinate relation, and from PTC's transformed signal, product identity, availability, producer-local validity, and MAP-facing coefficient. SCI-MAP does not repair or reinterpret those facts. VAL may register and evaluate a named rule, but it is not the author of producer or MAP policy.

Let i bind observation, detector occurrence and UID, stable RTC output sample n , exact PTC product/application generation, segment, and array/network/group; time is an attribute and storage positions are locators. Here $z_i = Z_i^{\text{PTC}}$ is the realized PTC transformed signal and γ_i is its exact PTC-owner-selected MAP-facing coefficient. The paired x/r symbols remain KID readout coordinates. The effective coefficient is $a_{pi} = G_{pi}\gamma_i$. The admitted `SCI-PTC:uniform_constant@draft-0.1` supplies exact dimensionless $\gamma_i = 1$, but neither Registry presence nor MAP permission selects or realizes it. Use requires explicit selection, complete atomic PTC publication, exact target/payload/occurrence association, a requested, applicable, eligible, realized `SCI-PTC:uniform_coefficient_handoff@draft-0.1` decision, independent MAP admission, and MAP finite-positive classification. An absent selection or incomplete or mismatched handoff makes the affected route unavailable; there is no unity, alternate-family, loading, sensitivity, scatter, or precision fallback.

The fixed contribution set $\mathcal{C}_p$ factors transformed-signal availability, PTC retention, coefficient availability, coefficient/QC, `SCI-MAP:map_upstream_admission@2`, AST validity, finite signal, finite positive coefficient, admitted projection/boundary, and MAP-local companion gates. Each retained typed state σ_g is projected to a Boolean membership predicate b_g ; T/F/U/C and VAL's four axes are never conjoined directly. Evaluation follows the structural, safe numerical-classification, coordinate/product, membership, and accumulation stages. Only structurally admitted payloads are read for MAP classification, and all required gates pass before payload arithmetic. Every source state, immutable reason, cause, provenance, and failure scope is retained. Let Π_{eff} be the recorded effective product support policy and $\mathcal{S}_{\text{out}}(\Pi_{\text{eff}})$ the exact set of map rows for which every support predicate that policy designates applicable passes. This is a row domain, not a numerical mask. The frozen ordinary result is then reused directly:

$$a_{pi} = G_{pi} \gamma_i, \quad (1)$$

$$\mathcal{G} = \{Z_{\text{avail}}, \text{PTC}_{\text{ret}}, \gamma_{\text{avail}}, \gamma_{\text{QC}}, \text{MAP}_{\text{adm}}, \text{AST}_{\text{coord}}, Z_{\text{finite}}, \gamma_{>0}, G_{\text{place}}, \text{MAP}_{\text{local}}\}, \quad (2)$$

$$b_g(i, p) = \text{pass}_g(\sigma_g(i, p)) \in \{0, 1\}, \quad g \in \mathcal{G}, \quad (3)$$

$$\text{pass}_{\gamma, \text{QC}}(i) = \mathbf{1} \begin{bmatrix} \text{request}_{\gamma, \text{QC}}(i) = \text{requested}, \\ \text{applicability}_{\gamma, \text{QC}}(i) = \text{applicable}, \\ \text{eligibility}_{\gamma, \text{QC}}(i) = \text{eligible}, \\ \text{realization}_{\gamma, \text{QC}}(i) = \text{realized} \end{bmatrix}, \quad (4)$$

$$\mathcal{C}_p = \left\{ i : \bigwedge_{g \in \mathcal{G}} b_g(i, p) = 1 \right\}, \quad (5)$$

$$\mathcal{S}_{\text{out}}(\Pi_{\text{eff}}) = \{p : \text{authorize}_{\Pi_{\text{eff}}}(p) = 1\}, \quad (6)$$

$$N_p = \sum_{i \in \mathcal{C}_p} a_{pi} z_i, \quad Q_p = \sum_{i \in \mathcal{C}_p} a_{pi}, \quad (7)$$

$$K_p = \sum_{i \in \mathcal{C}_p} a_{pi} k_i, \quad (8)$$

$$\hat{m}_p = N_p / Q_p, \quad \hat{k}_p = K_p / Q_p, \quad p \in \mathcal{S}_{\text{out}}, \quad (9)$$

The ten projections in Equation (3) are exact:

Pass predicate	Owner and object	Retained source codomain	Sole passing state	Nonpass and scope
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$b_{Z_{\text{avail}}}(i)$	PTC; transformed occurrence	signal availability with cause	exact available Z_i^{PTC}	unavailable, invalid, not requested, or conflict; occurrence/route exclusion
$b_{\text{PTC}_{\text{ret}}}(i)$	PTC policy and VAL; occurrence	named request, applicability, eligibility, and realization fields	requested, applicable, eligible , realized	every other field combination; occurrence/route scope
$b_{\gamma_{\text{avail}}}(i)$	PTC; coefficient slot/generation	family/generation, index/broadcast, product compatibility, readable payload, typed state/cause	exact compatible typed payload available; no finiteness claim	missing/unreadable, unavailable, conflict, or generation mismatch; route exclusion
$\text{pass}_{\gamma_{\text{QC}}}(i)$	PTC policy owner; coefficient occurrence	named request, applicability, eligibility, and realization fields plus causes	requested, applicable, eligible, and realized under the exact PTC profile	any other named-field combination is nonpassing; structural source-binding failure only when separately present; never MAP admission
$b_{\text{MAP}_{\text{adm}}}(i)$	MAP policy and VAL; occurrence	named request, applicability, eligibility, and realization fields	requested, applicable, eligible , realized	every other field combination; occurrence exclusion or source-binding block
$b_{\text{AST}_{\text{coord}}}(i)$	AST; same- n coordinate parent	availability, validity, frame, parent, cause	exact coordinate available and valid on matching parent	invalid, unavailable, conflict, or parent mismatch; occurrence exclusion
$b_{Z_{\text{finite}}}(i)$	MAP numerical gate; z_i	finite classification	finite	NaN, Inf, or unrepresentable; occurrence/product scope
$b_{\gamma>0}(i)$	MAP numerical gate; γ_i	positive; zero; negative; non-finite; unrepresentable class	finite and strictly positive	zero noncontributing; negative, non-finite, or unrepresentable invalid; occurrence scope
$b_{G_{\text{place}}}(i, p)$	MAP; occurrence-pixel relation	one-hot placement/boundary state	unique in-grid half-open containing pixel	invalid coordinate, outer-upper boundary, or out of grid; pixel nonmembership
$b_{\text{MAP}_{\text{local}}}(i, p)$	MAP; product row and role	local and companion decisions	every applicable local predicate passes	false, unknown, conflict, unavailable, or missing companion; declared scope

Boolean zero here means estimator nonmembership only. It is never multiplied into z_i , and it never erases the retained typed record or cause.

The supported division is not a recipe for filling unsupported pixels. Pre-normalized accumulators and representation-specific storage slots may exist outside $\mathcal{S}_{\text{out}}$, but the normalized scientific vector has no row there. A dense zero, sentinel, or coincidentally finite payload is not a semantic substitute and is not measured zero sky. Likewise, Q_p is a normalization fact. Its dimensions or historical name do not make it precision.

The active role is a calibrated- x -derived nonpolarimetric total-intensity-equivalent quantity in **mJy/beam**. The exact originating fixed-nominal-beam/template and CAL/PTC lineage are part of the quantity identity. Neither the unit nor a STOKES token promotes it to Stokes I. V0.2 does not authorize polarimetry, measured-R mapmaking, JINC, maximum-likelihood mapmaking, mosaicking, or reprojection.

2 Linear operator, response, and absolute mode

For the selected frozen plan Π , transformed-sample input domain $\mathcal{I}_\Pi$, projection matrix G , diagonal coefficient matrix Ω , neutral full-grid pixel vector $\mathbf{v}$, exact fixed-state upstream-to-PTC response $H_{\text{fixed},\Theta}$, and output-row selector J_{out} , the map operator, fixed-state response, and conditional covariance are:

$$D_Q = \text{diag}(G\Omega\mathbf{1}), \quad J_{\text{out}}\mathbf{v} = (v_p)_{p \in \mathcal{S}_{\text{out}}}, \quad D_{Q,\text{out}} = J_{\text{out}}D_QJ_{\text{out}}^\top, \quad (10)$$

$$A_{\text{MAP},\Pi} \equiv A_{\text{out}} \equiv D_{Q,\text{out}}^{-1}J_{\text{out}}G\Omega, \quad \widehat{\mathbf{m}}_{\text{out}} = A_{\text{MAP},\Pi}\mathbf{z}, \quad (11)$$

$$\delta\mathbf{m}_{\text{out}} = A_{\text{MAP},\Pi}H_{\text{fixed},\Theta}\delta\mathbf{s}, \quad R_{\text{fixed},\Theta} = A_{\text{MAP},\Pi}H_{\text{fixed},\Theta}, \quad C_{m,\text{out}} = A_{\text{MAP},\Pi}N_{\text{PTC}}A_{\text{MAP},\Pi}^\top. \quad (12)$$

The identity $A_{\text{MAP},\Pi} \equiv A_{\text{out}} \equiv D_{Q,\text{out}}^{-1}J_{\text{out}}G\Omega$ binds $\mathcal{I}_\Pi$, output-row domain $\mathcal{S}_{\text{out}}$, exact coefficient family/generation, `SCI-MAP:one_hot_containing_pixel01`, support policy, target WCS/grid, immutable parents, and lifecycle generation. The diagonal $D_{Q,\text{out}}$ contains only finite positive normalizations on $\mathcal{S}_{\text{out}}$, so its ordinary inverse is defined. The matrix has no rows outside that set. Extending it to the full storage grid by writing zeros would be a representation convenience, not this scientific operator, response, or covariance.

Every valid MAP product discloses this complete realized $A_{\text{MAP},\Pi}$ identity as its known MAP-local operator response. That disclosure is distinct from the upstream-conditioned response below and remains available when the latter is limited or unavailable.

Here N_{PTC} is a declared covariance of PTC-domain processed-sample error when such information is available, not the ensemble covariance of sky plus error. The state Θ contains all selected or estimated calibration, alignment, membership, coefficients, response, geometry, identity, and effective-support facts. Equation (12) therefore makes a conditional perturbation statement on exactly the authorized rows; it makes no unnamed μ_0 or residual-bias claim. If any element of Θ is estimated from the same data, an unconditional claim requires its joint uncertainty and cross-covariances.

The linear object $H_{\text{fixed},\Theta}$ has an exact source domain, PTC-sample codomain, unit transformation, parent, membership, support, coefficient state, and response basis. It is not a name for a refitted procedure. When available, $R_{\text{fixed},\Theta} = A_{\text{MAP},\Pi}H_{\text{fixed},\Theta}$ describes the response of that exact frozen state on the authorized output domain. A stored kernel image is the realized response to a named unit-bearing template $\boldsymbol{\tau}$ only if its sample-domain parent is $\mathbf{k} = H_{\text{fixed},\Theta}\boldsymbol{\tau}$ and the stored map is $A_{\text{MAP},\Pi}\mathbf{k} = R_{\text{fixed},\Theta}\boldsymbol{\tau}$ with the same exact row identity as signal. Without that identity it is a tracer, not the measured response. If any signal member lacks the required exact parent, response is limited or unavailable on the unchanged signal membership; no hidden subset, zero, or identity array can stand in for it. A realized PTC-grid companion begins at MAP and receives $A_{\text{MAP},\Pi}$ exactly once; it is never passed through upstream response again.

PTC full-procedure and PTC+MAP re-resolved responses are bounded finite-difference families, not one H matrix:

$$\Delta\mathbf{z}_{\text{PTC-FP}}(\delta\mathbf{s};\alpha) = \mathcal{P}_{\text{PTC}}(\mathbf{s} + \delta\mathbf{s};\alpha) - \mathcal{P}_{\text{PTC}}(\mathbf{s};\alpha), \quad (13)$$

$$\Delta\mathcal{S}_{\text{PTC-FP}} = \text{state}_{\text{PTC}}(\mathbf{s} + \delta\mathbf{s};\alpha) \ominus \text{state}_{\text{PTC}}(\mathbf{s};\alpha), \quad (14)$$

$$\Delta\mathbf{m}_{\text{PTC-FP}} = A_{\text{MAP},\Pi} \Delta\mathbf{z}_{\text{PTC-FP}}, \quad \text{only when MAP state } \Pi \text{ is held fixed}, \quad (15)$$

$$\Delta\mathbf{m}_{\text{PTC+MAP-RR}} = \mathcal{M}(\mathcal{P}_{\text{PTC}}(\mathbf{s} + \delta\mathbf{s};\alpha); \Pi_+) - \mathcal{M}(\mathcal{P}_{\text{PTC}}(\mathbf{s};\alpha); \Pi_0), \quad (16)$$

Here α binds amplitude, perturbation side, source state, location, domain, and every other family parameter. $\Delta\mathcal{S}_{\text{PTC-FP}}$ separately records changed masks, groups, rank, loading subspace, classifications, coefficient state, stop cause, support, and other selected state. Equation (15) applies only when MAP membership, coefficient, projection, and support remain fixed. Equation (16) instead uses separately resolved Π_0 and Π_+ ; it is the PTC+MAP re-resolved procedure response, not the fixed MAP operator applied to one matrix. The reserved whole-chain RTC-to-CAL-to-PTC-to-MAP response would rerun RTC, CAL, AST-dependent selection, PTC, MAP admission, coefficient/support resolution, and MAP from exact immutable parents under separate authority; it is unavailable unless separately established and is not claimed here. An unavailable upstream-conditioned response state does not invalidate the numerical map or prohibit later analysis. A later simulated response or response-corrected map is a new versioned product bound to the original MAP product and processing identity; it does not rewrite the original product's claim.

V0.2 fixes the signal-centering operator to $L = I$. SCI-MAP therefore does not remove a mean or a constant mode during coaddition. It preserves a constant admitted sample vector on the exact support-authorized output rows because numerator and denominator use identical coefficients. A stronger constant-sky statement additionally requires the forward operator to map that sky mode to the admitted sample constant. Neither result alone establishes point-source, integrated-flux, extended-source, edge, or absolute astrometric fidelity. The PTC nonrestored additive reference λ , fitted correlated removed component $\hat{U}_{\text{corr}}$, total removed component $U_{\text{total},\Theta}$, removed-subspace identity, fixed-state null space, full-procedure invariant/unidentifiable modes, and any separately authorized residual conditional bias are distinct facts. Any absolute mode already removed upstream remains absent and must be carried in response and null-mode provenance.

3 Input classes and falsifiable response behavior

The same response equation yields distinct, testable input classes without granting them equal scientific meaning:

Constant.

Fixed positive coefficients preserve a constant admitted sample vector even when coverage and variance vary.

Delta or basis pixel.

The output is a support-row-restricted column of $R_{\text{fixed},\Theta}$, including beam, projection, preprocessing, and boundary effects.

Point-source template.

The output is $R_{\text{fixed},\Theta}\tau_b$ on exactly the authorized rows. Peak, integrated, and fit amplitudes are different functionals.

Resolved, gradient, and Fourier structure.

The output is $R_{\text{fixed},\Theta}\tau$; constant preservation says nothing about these modes.

Variable coverage.

Fixed-coefficient normalization may preserve a constant while covariance and nonconstant response vary spatially.

Finite boundary.

Truncation can preserve a local constant yet change a compact-source peak or integral. No wrap or clamp is authorized.

The authorized base family is `SCI-MAP:one_hot_containing_pixel@1`. It uses half-open pixel cells, assigns unity to the unique finite in-grid containing pixel, excludes the outer upper boundary, never wraps or clamps, and has $\sum_p G_{pi} = 1$ only on that in-grid domain. Fractional projection remains a future role; no kernel, conservation, boundary-loss, response, or covariance semantics are inferred for it.

Response fidelity is therefore a validation layer, not a consequence of code executing the algebra. Delta, point-template, extended, Fourier, variable- coverage, and edge injections are preregistered in Appendix D; they are not run or claimed here.

4 Conditional uncertainty and statistical labels

For independent selected sample errors with an exact selected coefficient family proven to be true inverse variance, the general formal relation is:

$$\text{Var}(\hat{m}_p \mid \Theta) = \frac{\sum_i G_{pi}^2 \gamma_i}{(\sum_i G_{pi} \gamma_i)^2}, \quad \text{Cov}(\hat{m}_p, \hat{m}_r \mid \Theta) = \frac{\sum_i G_{pi} G_{ri} \gamma_i}{Q_p Q_r}, \quad p, r \in \mathcal{S}_{\text{out}}, \quad (17)$$

$$\phi_p = \frac{Q_p^2}{\sum_i G_{pi}^2 \gamma_i}, \quad \phi_p = Q_p \iff \sum_i \gamma_i (G_{pi}^2 - G_{pi}) = 0, \quad p \in \mathcal{S}_{\text{out}}, \quad (18)$$

Equation (18) states why normalization is not precision by default. One-hot geometry satisfies the projection part of the equality, but cannot establish that γ is calibrated inverse variance or that sample errors are independent. The admitted uniform family has exact value one but explicitly supplies no precision or independence meaning.

Four uncertainty objects must not be conflated:

1. Q , an ordinary normalization with its declared gridding unit;
2. $\phi = 1/[C_{m,\text{out}}]_{pp}$, a conditional marginal inverse variance under its named assumptions;
3. $C_{m,\text{out}}$ or $C_{m,\text{out}}^+$, the full conditional covariance or precision on the identifiable stochastic subspace of the exact authorized row domain; and
4. an empirical covariance, weight, or significance calibrated by NOI.

The bundle reports what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, lineage-resolvable, summarized, incomplete, or unavailable. Missing correlations are named, not set to zero. Neither normalization, hits, exposure, nor diagnostic weight is silently promoted to covariance. Absence of a complete covariance model does not invalidate the map or prohibit later scientific analysis. A later covariance estimate is a new versioned product bound to the original map and processing identity. The base map need not carry a complete covariance matrix or operator; it must disclose what actually persists and what remains lineage-resolvable or unavailable.

5 Distinct map facts, support, and MAP-local validity

The base/unfiltered MAP bundle separates facts because they answer different questions:

Fact	Meaning
AST-geometric incidence	Which coordinate-valid occurrences touch the unique containing pixel, independent of signal availability.
MAP-route-candidate incidence	Which exact realized PTC occurrences also pass the occurrence-level MAP upstream-admission profile.
Estimator-admitted contributions	How many occurrences passed every ordinary coefficient and contribution predicate.
Admitted observation count	How many complete observation bundles contribute to a coadd pixel; not sample hits or attempted observations.
Upstream-eligible original-footprint exposure	Unique original acquisition footprints ancestral to route candidates, placed by each original's own AST ALIGN-grid coordinate in the target WCS; not complete temporal support.
Retained original-footprint exposure	The same original-footprint estimand restricted to ancestors of estimator-admitted membership; not precision or effective integration time.
Normalization support	The numerical threshold predicate used to authorize ordinary division.
Science-policy support	The separately stricter adopted policy predicate.
MAP-local base-product validity	MAP's conjunction of admitted identity, support, finite signal, and no MAP-product failure; not downstream eligibility.

No one fact is an alias for another. In particular, exposure is not precision, positive normalization is not final validity, and a finite payload is not evidence of scientific availability.

ALIGN owns e_{acq} and valid-original e_{vo} , in seconds on stable original occurrences. RTC, CAL, and PTC preserve those facts, exact lineage, and causes. MAP deduplicates by observation, detector occurrence/UID, stable original/native occurrence, stable ALIGN slot, and ALIGN mapping generation. Under SCI-AST_TO_SCI-MAP_ORIGINAL_FOOTPRINT_COORDINATE v0.1/r0.1, each positive- e_{vo} original is placed exactly once using its layered AST ALIGN-grid direction/tangent/continuous-pixel parent in the exact target MAP WCS, with exact geometry/pointing realization, frame, validity, boundary, parent, and version. A descendant RTC-output coordinate is never substituted:

$$\mathcal{U}_p^{\text{up}} = \{u : e_u^{\text{vo}} > 0, G_p^u = 1, \exists i \in \mathcal{I}_{\text{route}} : u \rightsquigarrow i\}, \quad (19)$$

$$E_p^{\text{up,footprint}} = \sum_{u \in \mathcal{U}_p^{\text{up}}} e_u^{\text{vo}}, \quad (20)$$

$$\mathcal{U}_p^{\text{ret}} = \{u : e_u^{\text{vo}} > 0, G_p^u = 1, \exists i \in \mathcal{C} : u \rightsquigarrow i\}, \quad E_p^{\text{ret,footprint}} = \sum_{u \in \mathcal{U}_p^{\text{ret}}} e_u^{\text{vo}}, \quad (21)$$

$$\mathcal{U}_p^c = \bigcup_{o \in \mathcal{O}_p} \{(o, u) : u \in \mathcal{U}_{op}^{\text{ret}}\}, \quad E_p^c = \sum_{(o, u) \in \mathcal{U}_p^c} e_{ou}^{\text{vo}}. \quad (22)$$

The relation $u \rightsquigarrow i$ is causal influence only. One original may influence several RTC/PTC descendants at different MAP pixels, but those edges do not relocate or duplicate its physical detector-seconds. Under the base one-hot family an original contributes exposure to at most one pixel, or none after boundary loss. Upstream-eligible exposure uses originals ancestral to the exact route-candidate population; retained exposure restricts that population to ancestors of estimator-admitted outputs. Coadd identity includes the observation, so union never merges distinct acquisitions. Overlapping filters, decimation, replacement, synthesis, donor reuse, cadence, hits, Q , and γ create no exposure. Missing exact original lineage or own-coordinate authority makes the affected exposure role unavailable rather than reconstructible from duration or count. Causal influence/support is published separately from representative RTC occurrence and physical original-footprint exposure. Neither exposure plane is complete temporal support, effective map integration time, precision, or normalized-map influence.

The adopted v0.2 threshold rule is operational policy authority, not a derived physical law. For the declared plane and policy population of finite positive Q_p values, sorted in ascending order, the admitted formula names $c = \text{coverage_cut}$:

$$\mathcal{I}_{\Pi_{\text{eff}}} = \text{the declared support-policy population}, \quad (23)$$

$$\mathcal{P} = \{Q_p : \text{finite}(Q_p), Q_p > 0, p \in \mathcal{I}_{\Pi_{\text{eff}}}\}, \quad (24)$$

$$N = |\mathcal{P}|, \quad k = \left\lfloor \frac{\lfloor 0.75N \rfloor + N}{2} \right\rfloor, \quad (25)$$

$$Q_\star = \begin{cases} \mathcal{P}_{(k)}, & N > 0, \\ 0, & N = 0, \end{cases}$$

$$(c_{\text{req}}, c_{\text{eff}}, c_{\text{obs}}, c_{\text{real}}) = \text{requested, effective, observation-resolved, and realized exact states and values}, \quad (26)$$

$$c = c_{\text{real}} = \text{coverage_cut} \text{ (dimensionless scalar)}, \quad (27)$$

$$\text{admit}_{\Pi_{\text{eff}}}(c) = \mathbf{1}[\text{finite}(c), c \geq 0, c \leq 1 \vee \text{expert_request}(c)], \quad (28)$$

$$T_{\text{norm}} = Q_\star c / 10, \quad T_{\text{sci}} = Q_\star c, \quad (29)$$

$$S_p^{\text{norm}} = \mathbf{1}[\text{finite}(Q_p), Q_p > 0, Q_p \geq T_{\text{norm}}], \quad (30)$$

$$S_p^{\text{sci}} = \mathbf{1}[\text{finite}(Q_p), Q_p > 0, Q_p \geq T_{\text{sci}}]. \quad (31)$$

Both predicates require the pixel's own explicit finite positive coefficient; the empty-population threshold of zero does not manufacture support. The operator row set $\mathcal{S}_{\text{out}}$ is the exact intersection of the predicate row sets designated applicable by Π_{eff} ; it is not a full-grid mask with zero semantics.

The dimensionless $c = \text{coverage_cut}$ is finite and nonnegative. Exact zero requests no relative cut; both thresholds become zero, but the row's own finite strictly-positive normalization remains mandatory. Values $0 < c \leq 1$ are the ordinary recommended range. A value above one requires an explicit expert request and may honestly leave science support empty. A negative, non-finite, missing, or unrepresentable value fails before support rows are constructed or a required product is mutated. Requested, effective, observation-resolved, and realized exact state/value are retained. No universal numerical default is supplied. Both threshold comparisons are inclusive, and neither support state implies exposure, precision, significance, response, completeness, or physical optimality.

6 Observation coaddition on one common grid

Coaddition consumes immutable complete logical observation bundles, not independent planes. Physical persistence follows the effective output plan; conforming in-memory and persisted bundles have the same scientific identity, admission, and arithmetic. A coadd-only entry may consume persisted conforming bundles without rerunning upstream processing. Every required publication still completes or fails, and this rule does not waive upstream PTC completion. The MAP-owned aggregate profile `SCI-MAP:observation_coadd_admission@1` is registered and evaluated by SCI-VAL. Its four axes independently record request, applicability, eligibility, and realization. It atomically checks quantity and nominal-beam convention, PTC route/product role, response source/class, complete AST frame/WCS/grid, support policy, coefficient family, covariance disclosure, null/additive-reference state, required companions, lifecycle, and parentage before any coadd-owned state changes.

The canonical target grid may be requested and made immutable before observation-map construction. The only v0.2 placement is exact centered integer embedding on the same grid. Observation and coadd row and column differences are nonnegative and even; their half-differences are the integer offsets, and the shifted reference pixels identify the same world coordinate. The lossless typed WCS is the authority. A later crop is a newly identified row-selected product with an immutable parent and lineage; it cannot relabel that parent as compatible. No fractional shift, post-hoc pad, interpolation, reprojection, mosaicking, wrapping, or implicit source recentering is allowed.

For the atomically admitted per-pixel set $\mathcal{O}_p$, every contributing observation row is already a member of that observation's support-authorized scientific domain. Let $\mathcal{S}_{\text{out}}^c$ be the exact rows authorized by the effective coadd support policy. The MAP-owned family `SCI-MAP:uniform_observation_coadd_coefficient@1` assigns dimensionless $u_{op} = 1$ to every admitted row. It is equal-observation weighting, not precision. The coadd is:

$$\mathcal{S}_{\text{out}}^c = \{p : \text{authorize}_{\Pi_{\text{eff}}^c}(p) = 1\}, \quad (32)$$

$$u_{op} = 1 \quad \text{for each admitted observation row,} \quad (33)$$

$$N_p^c = \sum_{o \in \mathcal{O}_p} u_{op} \hat{m}_{op}, \quad Q_p^c = \sum_{o \in \mathcal{O}_p} u_{op}, \quad (34)$$

$$\hat{s}_p = N_p^c / Q_p^c, \quad p \in \mathcal{S}_{\text{out}}^c, \quad (35)$$

$$J_{\text{out}}^c \mathbf{v}^c = (v_p^c)_{p \in \mathcal{S}_{\text{out}}^c}, \quad (36)$$

$$\widehat{\mathbf{m}}_{\text{out}}^c = (\hat{s}_p)_{p \in \mathcal{S}_{\text{out}}^c} = B_{\text{out}} \widehat{\mathbf{m}}_{\text{obs}}, \quad (37)$$

$$\hat{k}_p^c = \frac{\sum_{o \in \mathcal{O}_p} u_{op} \hat{k}_{op}}{Q_p^c}, \quad p \in \mathcal{S}_{\text{out}}^c. \quad (38)$$

$$C_{c,\text{out}} = B_{\text{out}} C_{\text{obs}} B_{\text{out}}^\top \quad \text{only when every required block is available,} \quad R_{c,\text{out}} = B_{\text{out}} T_{\text{obs}}. \quad (39)$$

Here $\widehat{\mathbf{m}}_{\text{obs}}$ is the stacked signal on exact observation-row identities; B_{out} is the centered-integer equal-observation averaging operator; T_{obs} stacks exact compatible fixed-state observation response objects; and C_{obs} is the block covariance with every available within- and cross-observation block explicitly located. Equation (39) is a complete numerical covariance claim only when every block required by the exact covariance-qualified role is available. Otherwise the state is explicitly partial, symbolic, summarized, lineage-resolvable, or unavailable. Unknown blocks are not zero or independence. The coadd operator B_{out} has rows only in $\mathcal{S}_{\text{out}}^c$. Coadd accumulators or dense storage slots outside that set do not become a zero-valued scientific coadd. On its exact domain the coadd is a convex weighted mean. One admitted observation reproduces itself; two admitted observations produce their equal-observation mean. Its normalization is not precision. With cross-observation covariance, the exact conditional covariance retains those blocks on the same selected rows. A correlated-GLS benchmark may require negative combination coefficients, placing it outside this ordinary positive-coefficient contract.

The exact base-coadd role makes response advisory and unavailable-compatible; the response-bearing role requires compatible source domain, response basis, class, units, normalization, parent, and rows for every member. If any member response is unavailable or incompatible, the base role may retain the unchanged numerical signal membership but must report coadd response unavailable; it forms no hidden response subset. The covariance-qualified role similarly requires all named blocks. Retained exposure and admitted observation count follow the same atomic bundle admission, while exposure retains its unique-original own-coordinate placement. A rejected observation changes no coadd state anywhere, even if a subset of its planes would otherwise appear compatible.

7 Identity, WCS, units, and lifecycle

A map cannot be reconstructed from its collection position. Paired PTC x/r components remain upstream detector coordinates and are carried only when a MAP-facing fact requires their identity; MAP does not recast r as a response. The minimum identity includes the nonpolarimetric total-intensity-equivalent product role; external observation or coadd ID; array/group and occurrence domains; estimator, projection, PTC and coadd coefficient families; MAP-local operator and upstream-conditioned response/covariance states; signal unit and exact nominal-beam parent; complete AST frame/WCS; preconstruction target-grid identity where applicable; exposure convention; logical-bundle and persistence state; MAP profiles; exact PTC/AST generations and product identity. A STOKES field is only a locator unless separate quantity authority exists. The policy must be scientifically identifiable, durable, and comparable, but this contract prescribes no JSON, hash, or sidecar serialization. Array ID, name, index, network, map group, map index, detector occurrence, and container position remain distinct.

Science maps bind the complete frame identity supplied by `SCI-AST:rtc_output_grid_coordinates@1`, including the actual reference system, equinox/epoch, time scale, site/model state, TAN topology, WCS, axis order, signs, handedness, and degree-valued axes. MAP never equates J2000, FK5, ICRS, apparent, observed, or unrefracted frames by name/default. Pointing and OOF may reuse the exact ordinary MAP gridding operator when their inputs satisfy this contract. Their target WCS/frame, source model, fitting, response interpretation, outputs, defaults, profiles, and qualification remain mode- owned; no unsupplied mode profile becomes available. FITS/WCS coordinates are one-based and in-memory row/column indices are zero-based.

Full-precision typed or sidecar WCS is the lossless admission and provenance authority. A physical FITS serialization may differ by at most 0.1 arcsec in maximum sky separation while preserving exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations. This is a representation tolerance, not authority for a different grid.

Requested state records the immutable accepted intent. Effective state records the executable supported plan. Observation-resolved state replaces all observation-specific arrays, mappings, geometry, coefficients, and provenance. Realized state records what actually ran, all accumulators and distinct facts, products, offsets, representation states, cardinality, and failures. State flows one way; execution does not rewrite the request or inherit a prior observation's resolution.

8 Failure, downstream boundary, and immutable parentage

Failure semantics protect the scientific bundle from partial truth. An invalid candidate can be excluded at contribution scope; an unsupported pixel is MAP-local invalid; an incompatible observation is rejected atomically; an unrepresentable finite aggregate, coordinate/index conversion, missing required companion, or failed required publication propagates at its declared product scope before live state mutation. A completion marker cannot override a missing required output.

No downstream package receives a bare signal plane as a complete map. The boundary includes the estimator and normalization identity; response state; conditional covariance/formal-weight state and assumptions; distinct map facts; additive-bias and null-mode state; units; WCS; grouping; lifecycle; base/unfiltered MAP parentage; dependency versions; and failures. NOI may propagate a fixed operator but owns realization construction and empirical calibration. FLT owns filtered support, response, covariance, and validity. SRC, MODE, and BEAM own inference and fit uncertainty. FRUIT owns recurrence and iteration state.

Every later analysis is responsible for deciding which facts its own claim requires. An unavailable response prevents the original MAP bundle from making an unsupported response-dependent claim, but it does not prohibit analysis or invalidate other map claims. A significance analysis cannot relabel merely formal standardization; an absolute-level claim cannot ignore an unobservable offset. This contract defines no closed-world registry of all future uses. A downstream product may have local validity or eligibility, but it cannot rewrite MAP-local base-product validity or sever the immutable base/unfiltered MAP-parent identity.

When an admitted noise realization is described as using the same operator, membership, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and coadd admission are fixed or identically prescribed. Recomputing them from randomized data creates a different estimator, regardless of matching array shapes or filenames.

9 Uncertainty, validity, validation, and claim layers

The contract deliberately separates four reasoning layers:

1. **Uncertainty layer:** conditional covariance and response are propagated through the fixed operators; omitted state uncertainty remains explicit.
2. **Validity layer:** contribution eligibility, numerical support, science-policy support, and MAP-local base-product validity determine the exact rows on which the estimator is scientifically available; unsupported rows are absent, not zero-filled observations.
3. **Validation layer:** preregistered analytic fixtures, matrix comparisons, injections, covariance controls, representation round trips, and observational references test distinct claims.
4. **Readiness layer:** implementation evidence, operational reliability, data completeness, and release authority are assessed elsewhere.

Claim layer	Status here	Evidence needed before an achieved claim
Algebraic scientific contract	Proposed author draft	Scientific-owner review and contract freeze; analytic predictions are specified, not executed.
Engineering conformance	Not assessed	Requirement-by-requirement inspection and tests against one exact candidate identity.
Representation/response fidelity	Not assessed	WCS round trips and delta, template, extended, variable-coverage, and edge injections with preregistered bounds.
Observational performance	Not assessed	Independent astronomical amplitude, shape, position, covariance, and repeatability references where available.
Production readiness	Not assessed	Complete required artifacts, failure/log audit, operational gates, owner release decision, and no disqualifying gaps.

Floating reductions require a preregistered bound derived from operation count and conditioning. For a sum of n terms, a reference policy may use $\gamma_n = n\epsilon/(1 - n\epsilon)$ and a fixed multiple such as the admitted core's $8\gamma_n \sum_i |a_i|$, frozen before observing results. Integer counts, identities, masks, support, offsets, and product cardinality remain exact. Tolerances are not relaxed after failure.

The contract's 25 predictions are listed formally in Appendix D. No validation run, implementation inspection, observational reduction, or production decision was performed during this scientific authorship stage.

10 Owner decisions, external hard gates, and route disposition

All nine stable MAP decisions are decided for successor drafting. OD-001–007 and OD-009 use the approved 2026-08-28 dispositions; OD-008 retains the exact one-hot half-open projection. PTC-OD-010 supplies the versioned coefficient- Registry architecture, and the owner-approved uniform authority supplies one exact family, handoff profile, and separate MAP permission. These scientific decisions do not by themselves create an actual selection, publication, profile evaluation, application route, Registry activation, validation, or freeze.

Route level	Contract status	Exact remaining gate
Conditional estimator authority	Defined	Positive-coefficient observation and equal-observation centered-integer coadd equations are conditional scientific authority.
Source-closed numerical observation map	Conditionally defined	Requires an explicitly selected admitted family, complete PTC publication and exact handoff, admitted finite-nonnegative coverage_cut , every independent gate, and an exact successor source/profile binding. No realized application route is claimed.
Response-bearing science map	Conditional or unavailable	The MAP-local realized operator is disclosed; stronger claims require an exact sufficient fixed-state, PTC full-procedure, or PTC+MAP re-resolved family. Whole-chain RTC-to-CAL-to-PTC-to-MAP remains unavailable unless separately established.
Uncertainty-qualified map	Conditional or unavailable	Requires the exact covariance representation and every term/block needed by the named claim; unknown is not zero.
Compatible-grid coadd	Defined conditionally	Requires source-closed admitted observation bundles and observation_coadd_admission@1 ; response/covariance qualification remains role-specific.
Future reprojection/mosaic	Not authorized	A separate scientific contract remains required; preconstruction target-grid request and exact centered-integer embedding do not authorize resampling.

External hard gate	Contract disposition	Route effect
PTC MAP-facing coefficient	Uniform family admitted; actual handoff input-dependent	Registry presence alone realizes nothing. Missing selection, complete PTC publication, exact association, permission, or four-axis handoff blocks the affected route with no fallback.
Admitted numerical coverage_cut state	Scientific domain closed; actual state input-dependent	Bad or missing state fails before support construction or mutation; > 1 additionally requires an explicit expert request.
PTC-to-MAP uniform boundary	Approved exact uniform successor	The binding names the exact boundary ID and digest/version; a mismatch blocks the route.
MAP upstream-admission profile	Exact manifest-bound Registry record	A missing or incompatible SCI-MAP:map_upstream_admission@2 evaluation blocks candidate admission.
Exact AST same- <i>n</i> coordinate parent	Contract role closed; realization input-dependent	Missing/mismatched parent excludes the occurrence.

Original-footprint exposure lineage and AST ALIGN-grid parent	Exact boundary closed; realization input-dependent	Missing stable-original, target-WCS, validity, or boundary facts make required exposure unavailable and block a product that designates it required.
Response/covariance state	Claim-dependent	Does not by itself invalidate the base signal; blocks only a role whose exact policy requires stronger information.

The contract is coherent at this conditional boundary. A scientific route is defined only when every actual structural, selection, publication, handoff, admission, numerical, coordinate, support, and lifecycle gate passes. The candidate establishes no implementation conformity, response fidelity, validation, observational performance, freeze, readiness, Registry activation, application availability, or production authorization.

A Owner-decision register

This compact register is generated from `SCIENTIFIC_OWNER_DECISION_LEDGER.md`. The mechanical contract checker requires the nine exact IDs and decided dispositions to match in all views. Rendering this register does not freeze or activate the candidate.

Decision/status	Approved disposition	Binding consequence
SCI-MAP-0D-001 DECIDED	Relative support is an operational policy, with no exposure, precision, significance, response, completeness, or physical-optimality meaning.	Preserve the exact population, order statistic, thresholds, and policy identity.
SCI-MAP-0D-002 DECIDED	Independent experimentation is unrestricted; official identity and claims follow mathematical equivalence or versioned change control with appropriate evidence.	No permission process for research and no retroactive relabeling.
SCI-MAP-0D-003 DECIDED	Every valid map discloses its known MAP-local realized operator. Upstream- conditioned and whole-chain responses remain separately typed and honest.	Unsupported response-dependent claims remain unavailable; later estimates are new versioned products bound to the original.
SCI-MAP-0D-004 DECIDED	Report actual covariance information, domain, assumptions, omissions, limitations, and lineage; a complete covariance object is not mandatory.	Unknown is never zero or independence; later estimates have new identity.
SCI-MAP-0D-005 DECIDED	A complete logical observation bundle precedes coaddition; physical persistence follows the effective plan.	In-memory and persisted/coadd-only routes use identical admission, arithmetic, identity, and provenance; required publication still completes or fails.
SCI-MAP-0D-006 DECIDED	Pointing and OOF may reuse the exact ordinary MAP operator with conforming inputs.	Mode WCS/frame, model/fitting, interpretation, outputs, defaults, profiles, and qualification remain with their owners.
SCI-MAP-0D-007 DECIDED	<code>coverage_cut</code> is finite and nonnegative: zero gives no relative cut; $(0, 1]$ is ordinary; > 1 requires an explicit expert request; bad or missing values fail before support construction or mutation.	Retain requested, effective, observation-resolved, and realized states/values; no universal default.
SCI-MAP-0D-008 DECIDED	The one-hot lower-inclusive, upper-exclusive containing-pixel family remains the only ordinary projection.	Boundary loss contributes nowhere; fractional projection remains deferred.
SCI-MAP-0D-009 DECIDED	A canonical target grid may be requested before construction; compatible maps use exact centered-integer embedding.	A later crop is a new row-selected product; fractional resampling, reprojection, and mosaicking remain outside this contract.

B Requirement and crosswalk summary

The detailed 52-row requirement crosswalk and 25-row prediction crosswalk remain in `CROSSWALK.md`. This compact index makes the rationale's traceability visible without duplicating the normative clause text or the detailed row table. Appendix C renders every canonical requirement, and Appendix D renders every prediction, directly from the shared authority.

Authority range	Scientific surface	Principal rationale / remaining gate
REQ-001	Version and revision identity	Cover; owner freeze
REQ-002–010	Capability, input, identity, projection, coefficients, lifecycle, and admission	Sections 1, 7, 8; upstream producer facts
REQ-011–018	Ordinary accumulators, exact support-row publication, response, and absolute mode	Sections 1–3; actual response input
REQ-019–024	Row-selected covariance, formal weight, nuisance state, and statistical labels	Sections 2, 4; claim-specific covariance input
REQ-025–035	Separate incidence/contribution/exposure/support facts, threshold policy, exact rows, and MAP-local validity	Section 5; actual cut and support state
REQ-036–042	Complete bundles, atomic centered-integer coadd, row-selected coadd	Sections 6, 8; actual complete bundle and role-specific companions
REQ-043–048	response/covariance, and GLS boundary	
REQ-043–048	Product identity, WCS, provenance, failure, and publication	Sections 7–8; actual plan and mode/profile authority
REQ-049–052	Immutable base MAP parent, consumer boundary, fixed realization operator, and claim separation	Sections 8–9; consumer-specific claim gates
PRED-001–025	Analytic consequences and conformance/representation challenges	Sections 3, 9; Appendix D; not executed
OD-001–009	Stable scientific-owner decisions	Section 10; Appendix A; all nine DECIDED for successor drafting
SCI-MAP-CI-001	Resolved dimensionless classification and bounded amendment	Section 5; equations, REQ-031/032, and PRED-012 amended in place

C Canonical requirements

This appendix is rendered directly from the shared canonical LaTeX authority. Requirement identifiers are stable within SCI-MAP v0.2. Editing prose in this document does not silently change them.

SCI-MAP-REQ-001 – Versioned authority. The scientific contract version shall be v0.2. The rendered document revision shall be recorded separately as r0.1; changing layout or exposition alone shall not be represented as a new scientific contract version.

SCI-MAP-REQ-002 – Capability and realized PTC input. The ordinary input shall be the realized SCI-PTC v0.1/r0.5 transformed calibrated- x product $z_i = Z_i^{\text{PTC}}$, a nonpolarimetric total-intensity-equivalent detector-time quantity in the inherited top-of-atmosphere, point-source-equivalent mJy/beam convention with its exact originating fixed-nominal-beam identity. Neither that unit nor a STOKES token establishes Stokes I. MAP has no CAL-to-MAP fallback, no inferred no-op PTC product, and no product when the required positive-rank PTC product is absent or PTC is disabled.

SCI-MAP-REQ-003 – Exact occurrence identity. Every candidate occurrence i shall bind observation, detector occurrence and UID, stable RTC output sample n , exact PTC product/application generation, segment, and array/network/group as applicable. Sample time is an associated attribute. Input-column, row, and container positions are locators only and shall not establish identity or equality.

SCI-MAP-REQ-004 – Producer facts and MAP admission. SCI-MAP shall consume PTC-owned transformed-signal availability, output-retention disposition, coefficient family/value/typed availability, and coefficient/QC proposition without reinterpretation. The PTC coefficient/QC proposition passes only when the coefficient is requested, applicable, eligible, and realized under the exact PTC-owned coefficient/QC profile for the declared MAP-facing family. It does not grant MAP use. SCI-MAP shall separately apply the source-bound occurrence profile SCI-MAP:map_upstream_admission@2; the manifest-bound SCI-VAL Registry revision binds and VAL Core evaluates that MAP-authored policy but does not author it. The immutable r0.5 profile SCI-MAP:map_upstream_admission@1 is historical, is not a compatibility alias, and shall not replace SCI-MAP:map_upstream_admission@2. Every fact, decision, cause, scope, and immutable reason remains separately typed.

SCI-MAP-REQ-005 – Coordinate and projection authority. MAP shall consume SCI-AST:rtc_output_grid_coordinates@1 for the same stable RTC sample n and complete parent chain as z_i ; it shall never join by time, row, cardinality, or coordinate equality. MAP owns G_{pi} through SCI-MAP:one_hot_containing_pixel@1: the unique finite in-grid pixel whose cell contains the coordinate under half-open bounds contributes $G_{pi} = 1$, every other pixel has $G_{pi} = 0$, and an outer-upper-boundary or out-of-grid coordinate contributes nowhere. Fractional projection is deferred. AST may materialize the relation only when the artifact binds the AST parent and exact MAP plan.

SCI-MAP-REQ-006 – Ordinary coefficient status. The ordinary route requires one explicitly selected PTC-owned MAP-facing analysis/gridding coefficient family satisfying SCI-PTC-REQ-052–055 and SCI-PTC_TO_SCI-MAP_BOUNDARY v0.2-draft.1/r0.4. The admitted Registry SCI-PTC:analysis_gridding_coefficients@draft-0.1 contains SCI-PTC:uniform_constant@draft-0.1, whose exact supported value is dimensionless $\gamma_i = \omega_i = 1$, and grants MAP permission through SCI-PTC:uniform_coefficient_handoff@draft-0.1. Registration and permission do not select a family or realize a handoff. MAP shall use this family only after explicit selection, complete atomic PTC publication q , exact target/payload/occurrence association, and a requested, applicable, eligible, realized profile evaluation under the admitted source and compatible binding generation. MAP does not reproduce, select, or infer the PTC-owned formula. An absent selection without an authorized mode default, unregistered family, incomplete PTC publication, missing or mismatched payload, unavailable coefficient/QC decision, or incompatible generation makes the affected route unavailable; there is no unity or alternate-family fallback. Structural availability remains distinct from finiteness. MAP alone classifies an authorized payload as finite positive, exact zero, negative finite, non-finite, or unrepresentable and applies only a matching finite strictly-positive value. The uniform family supplies no precision, sensitivity, loading, scatter, response, covariance, exposure, validity, optimality, or default meaning.

SCI-MAP-REQ-007 – Exposure lineage and accounting. ALIGN-owned e_{acq} and valid-original e_{vo} , in seconds, shall remain immutable on stable original occurrences through RTC/CAL/PTC. The products are normatively `upstream_eligible_original_footprint_exposure` and `retained_original_footprint_exposure`. The shorter prose labels “upstream-eligible exposure” and “retained exposure” are explicit aliases for those

products and no others. MAP shall deduplicate by exact observation, detector occurrence/UID, stable original/native occurrence, stable ALIGN slot, and ALIGN mapping generation; place each positive- e_{vo} original exactly once through SCI-AST_TO_SCI-MAP_ORIGINAL_FOOTPRINT_COORDINATE v0.1/r0.1, its layered AST ALIGN-grid direction/tangent/continuous-pixel parent in the exact target MAP WCS, and SCI-MAP:one_hot_containing_pixel01; and never use a descendant RTC-output coordinate. Geometry/pointing realization, frame, coordinate validity, boundary state, parent, and version are required. Upstream-eligible and retained exposure differ only by ancestor-population restriction. Causal influence, representative RTC occurrence, and original-footprint exposure remain separate. These exposure products are not complete temporal support, effective map integration time, precision, or normalized-map influence. Missing original-coordinate authority makes the affected exposure role unavailable. Invalid-original has $e_{vo} = 0$; synthesis, replacement, donor reuse, overlapping filters, and decimation create no exposure. Boundary loss contributes nowhere. Coadd exposure unions atomically admitted observation-scoped original identities.

SCI-MAP-REQ-008 – Response information state. Every valid MAP product shall disclose its known MAP-local realized operator response $A_{MAP,\Pi}$, including exact input/output domains, membership, projection, coefficient, support-row, WCS, parent, lifecycle, and limitations. MAP shall separately type upstream-conditioned fixed-state linear response, PTC full-procedure finite differences, and PTC+MAP re-resolved procedure response. An available fixed-state object shall also bind source domain, PTC-sample codomain, units, parents, response basis, and upstream limitations; a full-procedure object shall additionally bind perturbation family α and a separately typed ΔS_{PTC-FP} state-change record. The PTC+MAP family reruns PTC and resolves Π_0 and Π_+ . A whole-chain RTC-to-CAL-to-PTC-to-MAP response is reserved for a separately authorized complete study and is unavailable unless so established. Absent upstream response shall be reported as limited or unavailable without concealing the known MAP-local operator, invalidating an otherwise valid base map, fabricating a whole-chain identity, or prohibiting later analysis. A later response estimate or corrected map is a new immutable versioned product bound to the original MAP product and processing identity.

SCI-MAP-REQ-009 – Immutable plan and lifecycle. The accepted request shall be immutable and shall resolve one way through requested, effective, observation-resolved, and realized states. Disabled expert values remain inactive; one observation's resolved state shall not leak into another.

SCI-MAP-REQ-010 – Factored contribution predicate. For each pixel, membership in $\mathcal{C}_p$ shall be the conjunction of ten exact Boolean pass predicates derived from separately retained typed records: transformed-signal availability; PTC output retention; coefficient availability; PTC coefficient/QC proposition pass $_{\gamma,QC}$; MAP upstream admission; AST coordinate validity; finite z_i ; finite strictly positive γ_i ; admitted projection/boundary relation; and MAP-local contribution plus required-companion predicates. Every VAL-evaluated gate retains named request, applicability, eligibility, and realization fields; only request=**requested**, applicability=**applicable**, eligibility=**eligible**, and realization=**realized** passes. Every other field combination is nonpassing with its exact cause, failure scope, and immutable reason retained. A structural source- or profile-binding failure is recorded only when that structural condition actually occurs; an ordinary nonpassing field value is not thereby a structural failure. Evaluation follows the ordered DAG: A structural identity/parent, generation, source/profile, producer availability, and decision-realization checks; B retrieval only of structurally admitted payloads and MAP classification of z_i and γ_i ; C AST validity, one-hot boundary/placement, and MAP-local required companions; D membership construction; E accumulation. All gates pass before payload arithmetic. Reading an authorized payload to classify it is not arithmetic, and excluded non-finite values never contaminate accumulation. T/F/U/C or four-field records shall never be conjoined directly; no numeric-zero mask represents exclusion.

SCI-MAP-REQ-011 – Ordinary accumulators. On the fixed contribution set, SCI-MAP shall accumulate $N_p = \sum_i a_{pi} z_i$, $Q_p = \sum_i a_{pi}$, and, when available for the identical membership, $K_p = \sum_i a_{pi} k_i$, where $a_{pi} = G_{pi} \gamma_i > 0$.

SCI-MAP-REQ-012 – Normalized publication. The scientific normalized output vector shall contain exactly the rows $p \in \mathcal{S}_{out}$ authorized by the applicable effective support policy. On those rows finite $Q_p > 0$ gives $\hat{m}_p = N_p/Q_p$, and an available response companion gives $\hat{k}_p = K_p/Q_p$. No unsupported row, zero-filled value, or sentinel is a normalized scientific output or measured zero.

SCI-MAP-REQ-013 – Accumulator interpretation. The numerator is a pre-normalized accumulator and Q_p is a normalization fact by default. Neither shall be labeled a sky estimator, inverse variance, confidence, coverage, or significance without the separately required conditions.

SCI-MAP-REQ-014 – Constant preservation. With fixed admitted membership and positive coefficients, the ordinary estimator shall preserve a constant admitted sample input on every row in its exact support-authorized output set. This prediction shall not create rows outside that set or be promoted to compact-source, extended-source, integrated-flux, absolute-offset, or astrometric fidelity.

SCI-MAP-REQ-015 – Projection-specific estimand. The authorized base projection is exactly `SCI-MAP:one_hot_containing_pixel01`; it preserves unity over the finite in-grid domain and has explicit half-open boundary loss. Fractional projection, diagonal weighted least squares, and conservation across a finite boundary are outside this branch until separately authorized.

SCI-MAP-REQ-016 – One exact MAP operator for linear companions. For one selected immutable realized plan Π , $A_{\text{MAP},\Pi} \equiv A_{\text{out}} \equiv D_{Q,\text{out}}^{-1} J_{\text{out}} G \Omega$ is the only MAP operator and its complete realized identity is the MAP-local operator-response disclosure. Its identity binds the exact admitted PTC-sample input domain, support-authorized output-row domain, coefficient family/generation, one-hot projection plan, support policy, target WCS/grid, immutable parents, and lifecycle generation. Signal, fixed-state response, conditional covariance, an available realized PTC-grid response companion, fixed noise realizations, and a PTC full-procedure finite difference with MAP state fixed shall use that same operator, membership, placement, coefficient state, and output rows. A PTC-grid companion begins at MAP and receives the operator exactly once, never $H_{\text{fixed},\Theta}$ again. Physical original-footprint exposure instead uses unique originals and each original's own AST ALIGN-grid coordinate; causal influence is not a linear exposure companion.

SCI-MAP-REQ-017 – Realized response claim. A stored kernel may be called a realized fixed-state response to template τ only when its exact sample parent satisfies $k = H_{\text{fixed},\Theta} \tau$ and the support-restricted map satisfies $\hat{k}_{\text{out}} = A_{\text{MAP},\Pi} H_{\text{fixed},\Theta} \tau = R_{\text{fixed},\Theta} \tau$ on exactly the signal rows. A PTC full-procedure finite difference $\Delta z_{\text{PTC-FP}}$ plus its separate state-change record, and a PTC+MAP re-resolved difference shall retain their distinct family/domain identities and shall not be represented by that fixed matrix. Otherwise the response is a tracer, limited, or unavailable.

SCI-MAP-REQ-018 – Absolute-offset mode. SCI-MAP v0.2 shall apply no signal-centering operation: $L = I$. Coaddition shall not subtract a mean, remove a constant mode, or recenter a source; any upstream loss of that mode shall remain explicit in response and provenance.

SCI-MAP-REQ-019 – Conditional covariance. When a declared PTC-domain covariance N_{PTC} is available for the exact fixed membership and state Θ , MAP shall propagate it as $C_{m,\text{out}} = A_{\text{MAP},\Pi} N_{\text{PTC}} A_{\text{MAP},\Pi}^T$, retaining consequential off-diagonal terms on the exact operator domains. The equation is complete only on the declared available domain. Missing, limited, symbolic, summarized, lineage-resolvable, or unavailable terms remain typed and are never zero-filled or interpreted as independence; absence of a complete model does not invalidate the signal map.

SCI-MAP-REQ-020 – When normalization is precision. Normalization may be called marginal inverse variance only when the exact selected γ family is calibrated true inverse variance, selected errors are independent on the asserted domain, and $\sum_i \gamma_i (G_{pi}^2 - G_{pi}) = 0$. One-hot geometry alone does not prove the coefficient premise.

SCI-MAP-REQ-021 – Formal diagonal weight. When its assumptions hold, $\phi_p = Q_p^2 / \sum_i G_{pi}^2 \gamma_i$ is the formal marginal inverse variance. It remains distinct from Q_p , support-restricted precision $C_{m,\text{out}}^+$, exposure, hits, and empirical statistical weight.

SCI-MAP-REQ-022 – Covariance and nuisance disclosure. The bundle shall report what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, lineage-resolvable, summarized, incomplete, or unavailable. It shall list omitted correlations and nuisance terms; absence of a complete covariance model shall not invalidate the map or prohibit later analysis, and unavailable information shall not be set to zero. A later covariance estimate may be attached only as a new versioned product bound to the original MAP product and processing identity.

SCI-MAP-REQ-023 – Data-derived state. If membership, coefficients, response, or geometry are estimated or selected from the same data, the displayed linear expectation and covariance shall remain explicitly conditional; unconditional bias or covariance requires a named joint model or resampling result outside this fixed-state claim.

SCI-MAP-REQ-024 – No significance promotion. Signal multiplied by the square root of merely formal weight shall be labeled formal standardized signal, not `sig2noise` or empirical significance. Empirical calibration

remains NOI-owned.

SCI-MAP-REQ-025 – Distinct map facts. Every base/unfiltered MAP bundle shall separately represent AST-geometric incidence, MAP-route-candidate incidence, estimator-admitted contribution, admitted-observation count, `upstream_eligible_original_footprint_exposure`, `retained_original_footprint_exposure`, numerical-normalization support, science-policy support, and MAP-local base-product scientific validity. No fact or downstream eligibility decision shall become authority for another.

SCI-MAP-REQ-026 – Incidence layers. AST-geometric incidence records valid coordinate placement independent of signal availability; MAP-route-candidate incidence additionally requires the exact realized PTC route and occurrence-level upstream admission; estimator contribution additionally requires all remaining gates in C_p . The one-hot touch convention counts the unique containing pixel only.

SCI-MAP-REQ-027 – Estimator contributions. Estimator-admitted contributions shall count exactly $i \in C_p$. A candidate ineligible under the exact MAP admission or contribution policy for its preserved cause, or with unavailable/conflicting required state, zero/negative/non-finite coefficient, non-finite signal, or out-of-grid projection, contributes nowhere; no producer flag implies a universal action.

SCI-MAP-REQ-028 – Admitted observation count. For coadds, admitted observation count shall count complete observation bundles contributing at the pixel and shall not be confused with sample hits or observation attempts.

SCI-MAP-REQ-029 – Upstream-eligible original-footprint exposure. MAP `upstream_eligible_original_footprint_exposure` shall union and deduplicate exact positive- e_{vo} original occurrences that are ancestors of route candidates admitted by `SCI-MAP:map_upstream_admission@2`. Each original is placed once by its own exact AST coordinate and the one-hot half-open rule; it is not placed at every descendant coordinate. The exposure population precedes coefficient positivity and MAP-local numerical rejection. Missing original-coordinate or lineage authority makes this exposure role unavailable.

SCI-MAP-REQ-030 – Retained original-footprint exposure. MAP `retained_original_footprint_exposure` shall use the same original-occurrence key and own-coordinate placement as REQ-029, restricted to originals that are ancestors of estimator-admitted members. Overlapping filters, decimation, replacement, synthesis, and multiple descendants do not duplicate seconds. Coadded exposure shall union observation-scoped original identities after atomic coadd admission; one original can occupy at most one pixel under the base one-hot family, or none after boundary loss. Neither exposure is precision.

SCI-MAP-REQ-031 – Adopted threshold selector. For each declared support-policy population, sort its finite strictly positive normalization coefficients Q_p ; for count $N > 0$, select zero-based $k = \lfloor ([0.75N] + N)/2 \rfloor$ and $Q_\star = Q_{(k)}$; for empty input set $Q_\star = 0$. The exact population and policy identities and the requested, effective, observation-resolved, and realized state/value of dimensionless $c = \text{coverage_cut}$ shall be retained. The scalar shall be finite and nonnegative. Exact zero requests no relative cut, while finite strictly-positive Q_p remains mandatory. Values $0 < c \leq 1$ are the ordinary recommended range. A value $c > 1$ is admissible only as an explicit expert request and may honestly produce empty science support. Negative, non-finite, missing, or unrepresentable values shall fail before support-row construction or required-product mutation. No universal numerical default is established.

SCI-MAP-REQ-032 – Separate support predicates. Only after the effective support policy admits the exact finite nonnegative c under REQ-031, numerical normalization support shall require finite positive $Q_p \geq Q_\star c/10$, and science-policy support shall independently require finite positive $Q_p \geq Q_\star c$. Both comparisons are inclusive and use the exact declared population and inherited order statistic. At $c = 0$, no relative threshold excludes a row, but no zero, negative, non-finite, or absent Q_p is restored. A permitted expert $c > 1$ may leave the science-support set empty. These thresholds are an operational relative-support policy and imply no exposure, precision, significance, response, completeness, or physical optimality. Neither support state establishes final validity or precision.

SCI-MAP-REQ-033 – MAP-local base-product validity. One MAP-local base-product scientific-validity state shall require membership in the exact support-authorized row set, exact admitted PTC and AST parents, passing MAP contribution/support policy, finite normalized signal, and absence of an applicable MAP-product failure. Explicitly limited or unavailable response/covariance does not alone invalidate the map. Retained storage, finite payload, coefficient, exposure, incidence, normalization, or downstream-local eligibility cannot establish or rewrite MAP validity.

SCI-MAP-REQ-034 – Classify before arithmetic. Only structurally admitted payloads shall be retrieved and safely classified for finiteness/sign before membership construction. All required gates shall pass before payload arithmetic. Multiplication by zero, masking after accumulation, or sanitizing a non-finite aggregate shall not substitute for admission.

SCI-MAP-REQ-035 – Exceptional contribution states. Zero coefficient shall be noncontributing, a low positive coefficient shall follow the explicit adopted policy, negative or non-finite ordinary coefficients shall be invalid, and an out-of-bounds projection shall contribute nowhere. Each cause shall remain distinguishable.

SCI-MAP-REQ-036 – Complete observation bundle. A base/unfiltered observation map shall exist as one immutable ordered logical bundle before it can participate in coaddition. The bundle contains signal, numerator, normalization identity, MAP-local realized operator response, upstream-conditioned response class/state, uncertainty/covariance state, the distinct facts in REQ-025, exposure lineage/accounting identity, unit and exact nominal-beam parent, complete AST frame/WCS, shape/indexing, estimator/projection/profile identities, PTC removed-component/additive-reference/null-space state, lifecycle, exact PTC/AST parentage, and product identity. Physical persistence is separately controlled by the effective output plan. Limited or unavailable upstream response/covariance states are complete disclosures, not automatic signal invalidity.

SCI-MAP-REQ-037 – Atomic coadd admission. A complete observation bundle shall pass all compatibility and required-companion checks before any coadd-owned state changes. Rejection shall leave numerators, normalizations, counts, exposure, response, validity, provenance, and product cardinality unchanged.

SCI-MAP-REQ-038 – Centered integer common grid. A canonical target grid may be requested and made immutable before observation-map construction. Observation maps constructed on that exact grid, or otherwise compatible observation/coadd shapes, shall use exact centered-integer embedding: shape differences are nonnegative and even in each axis, and the corresponding integer offset maps reference pixels to the same world coordinate. A later crop is a newly identified row-selected product with immutable parent and lineage; it shall not relabel the parent or make it retrospectively compatible. Fractional shift, post-hoc pad, reprojection, interpolation, wrapping, implicit recentering, mosaicking, and a different sky grid are outside this estimator.

SCI-MAP-REQ-039 – Ordinary coadd estimator. The exact v0.2 family `SCI-MAP:uniform_observation_coadd_coefficient@1`, owned by MAP, assigns dimensionless $u_{op} = 1$ to every row of each bundle eligible and realized under the VAL-governed aggregate profile `SCI-MAP:observation_coadd_admission@1`. It is an equal-observation coefficient, not precision or inverse variance. On exact centered-integer row identities, the coadd accumulates $N_p^c = \sum_o u_{op} \hat{m}_{op}$, $Q_p^c = \sum_o u_{op}$, and publishes $\hat{s}_p = N_p^c / Q_p^c$ only on $\mathcal{S}_{out}^c$. Conforming in-memory and persisted bundles have identical admission, identities, arithmetic, and result; storage route is not a scientific weighting or eligibility fact.

SCI-MAP-REQ-040 – Coadd companions and unavailable response. The stacked signal vector $\hat{\mathbf{m}}_{obs}$ shall carry explicit observation-row identities; B_{out} is the exact centered-integer equal-observation averaging operator; and T_{obs} is the stacked compatible fixed-state observation-response object on those same domains. Under the exact response-independent base-coadd role, response is advisory and unavailable-compatible: if any admitted member response is unavailable or basis-incompatible, the unchanged numerical signal membership may coadd only with coadd response typed unavailable, never a hidden subset or zero companion. A response-bearing coadd role instead requires every member response and exact compatible source-domain, basis, class, units, normalization, and row identity. Exposure and count follow the same atomic bundle admission, with exposure's unique-original placement.

SCI-MAP-REQ-041 – Coadd covariance and precision. C_{obs} shall be the block matrix on exact stacked observation-row identities with every available within- and cross-observation block explicitly located. Only when every block required by the exact covariance-qualified role is available may MAP publish the complete numerical equation $C_{c,out} = B_{out} C_{obs} B_{out}^T$. Otherwise covariance shall be explicitly partial, symbolic, summarized, lineage-resolvable, or unavailable; missing blocks are unknown, never zero or independence. The base signal coadd may remain valid under its named unavailable-compatible role. The uniform observation coefficient is not precision.

SCI-MAP-REQ-042 – Correlated-GLS boundary. A correlated minimum-variance solution using $K_p^{-1} \mathbf{1}$ may require negative combination coefficients and therefore belongs to a separately authorized correlated-GLS branch, not the ordinary positive-coefficient coadd.

SCI-MAP-REQ-043 – Minimum product identity. Every observation and coadd product shall identify its nonpolarimetric total-intensity-equivalent role; external observation/coadd, array/group, and occurrence domains;

estimator and coefficient families; MAP-local operator and upstream-conditioned response/covariance states; signal unit and fixed-nominal-beam parent; complete AST frame/WCS; preconstruction target-grid request where applicable; grid/projection/boundary; MAP profiles; exact PTC product/application and coefficient generations; exposure convention; logical-bundle and plan-controlled persistence state; lifecycle; parentage; causes; and product identity. No STOKES token supplies quantity authority. Scientific identity shall be durable without prescribing a persistence format.

SCI-MAP-REQ-044 – Frames and units. MAP shall bind the complete frame identity delivered by `SCI-AST:rtc_output_grid_coordinates@1`: exact source/target frame, reference system (including ICRS or FK5/J2000 as actually declared), equinox/epoch, time scale, site/model state, TAN topology, WCS, axis order, signs, handedness, and degree-valued science axes. MAP shall not equate J2000, FK5, ICRS, apparent, observed, or unrefracted frames by name or default. Signal/response units are `mJy/beam` with exact nominal-beam identity; covariance has its square.

SCI-MAP-REQ-045 – WCS and index persistence. Full-precision typed or sidecar WCS shall be the lossless admission and provenance authority. A physical FITS WCS may differ by at most 0.1 arcsec maximum sky separation while preserving exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations. FITS coordinates are one-based and memory indices zero-based.

SCI-MAP-REQ-046 – Realized provenance. Realized state shall record requested, effective, observation-resolved, applied, and realized identities, including all four exact `coverage_cut` stages; exact PTC/AST/boundary/profile and admitted coefficient Registry/family/handoff generations; every typed gate's named request, applicability, eligibility, and realization fields plus Boolean projection, immutable reason, cause, failure scope, and DAG stage; admitted contributions/bundles; the complete MAP-local realized operator $A_{\text{MAP},\Pi}$, accumulators, and row domains; unique-original exposure keys, stable ALIGN slots, layered AST ALIGN-grid coordinate parents, preconstruction target grid, boundary states, and separate influence edges; PTC λ , $\hat{U}_{\text{corr}}$, $U_{\text{total},\Theta}$, removed-subspace and null-space identities; fixed-state, PTC full-procedure, and PTC+MAP re-resolved response family and state-change record; covariance state; centered-integer offsets; logical-bundle and persistence-plan state; required products; and failures.

SCI-MAP-REQ-047 – Failure scope and atomicity. An invalid required input, incompatible bundle, unrepresentable finite aggregate, unrepresentable coordinate/index conversion, missing required companion, or failed required publication shall propagate at its declared contribution, pixel, observation-admission, or complete-product scope and shall fail before live aggregate state changes.

SCI-MAP-REQ-048 – Required observation products. A complete logical observation bundle shall exist before coaddition. Persisted per-observation products are the normal scientific expectation and shall be published whenever the effective v0.2 output plan designates them required; every required publication shall complete or the declared operation fails without a completion claim. A plan may instead permit a fully in-memory route. A coadd-only entry may consume persisted conforming observation bundles without rerunning upstream processing. In-memory and persisted routes shall use identical logical admission, arithmetic, product identity, and provenance. Optional nonpublication does not imply that the logical observation product did not exist, and this MAP rule does not waive PTC's upstream atomic required-publication obligations.

SCI-MAP-REQ-049 – Immutable base parent. Downstream transformations shall preserve resolvable base/unfiltered MAP-bundle identity and MAP-local base-product validity. A later local validity/eligibility decision may differ for its named use but shall not rewrite MAP validity, alter the original claims, or sever parentage. Later response or covariance estimates are new versioned products.

SCI-MAP-REQ-050 – Complete consumer boundary. No consumer shall receive a bare signal plane as a scientifically complete map. The boundary shall include quantity and nominal-beam identity, estimator/normalization/coefficient/profile identity, honest response class/state and uncertainty/covariance disclosure, distinct map and exposure facts, PTC additive-reference/removed-component/null-space state, exact frame/WCS, shape/indexing, lifecycle, parentage, dependency versions, and failures. Later analyses choose their own claim requirements and may attach honest versioned response/covariance products; MAP does not own their eligibility.

SCI-MAP-REQ-051 – Same fixed noise operator. An admitted noise realization may be described as using the same map/coadd operator only when eligibility, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and atomic coadd admission are fixed or identically prescribed. Re-estimation from the randomized data defines a different named estimator.

SCI-MAP-REQ-052 – Method and claim separation. Pointing and OOF may be registered consumers of the exact ordinary MAP gridding operator only when their inputs satisfy this operator contract. Reuse grants no mode-specific WCS/frame, source model, fitting, response interpretation, output, default, or qualification authority; an unsupplied mode profile remains unavailable. This authority shall not otherwise be applied by analogy to JINC, maximum-likelihood mapmaking, OOF residual transfer, reprojection, filtering, empirical noise, fitting, Beammap inference, feedback, Stokes Q/U, measured-R mapmaking, or other units. Independent inspection, modification, experimentation, and proposal remain unrestricted, while an altered scientific identity or official claim requires an explicit versioned successor and evidence appropriate to that claim. Algebraic contract correctness, engineering conformance, representation/response fidelity, observational performance, and production readiness shall remain separate claims.

D Falsifiable predictions

These predictions are preregistered contract consequences or conformance challenges. Their presence is not validation evidence.

SCI-MAP-PRED-001 – Uniform constant. For fixed membership, a constant admitted input with uniform positive coefficients produces that constant on every support-authorized MAP-valid output row; numerator and normalization scale together, and no unsupported scientific row is created. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-002 – Unequally weighted constant. For fixed membership, unequal positive coefficients still preserve the same constant on every support-authorized output row, while normalization and conditional variance generally change with position; rows outside that set have no normalized scientific value. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-003 – One-pixel hand calculation. For an authorized row with admitted z_i and $a_i > 0$, the exact triplet is $(N, Q, \hat{m}) = (\sum a_i z_i, \sum a_i, \sum a_i z_i / \sum a_i)$; an unauthorized row has no normalized scientific value. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-004 – One-hot boundary. A finite coordinate inside exactly one half-open cell has one $G_{pi} = 1$, preserves unity over pixels, and contributes only there. A coordinate on an internal upper edge belongs to the adjacent cell; a coordinate on the outer upper extent or outside the grid contributes nowhere. Fractional splitting is not authorized. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-005 – Delta input. For a declared fixed state, a basis-pixel sky perturbation produces the corresponding support-row-restricted column of $R_{\text{fixed}, \Theta} = A_{\text{MAP}, \Pi} H_{\text{fixed}, \Theta}$, including projection and boundary truncation; a PTC full-procedure or PTC+MAP re-resolved difference is separately typed and need not equal that column. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-006 – Point-source template. A declared point-source perturbation τ_b produces $R_{\text{fixed}, \Theta} \tau_b$ only for the exact fixed response state and authorized rows; PTC full-procedure and PTC+MAP re-resolved families retain their perturbation/state parameters. The unavailable whole-chain RTC-to-CAL-to-PTC-to-MAP family is not inferred. Peak, integrated, and fitted amplitudes remain distinct functionals. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-007 – Resolved and Fourier inputs. At fixed state, resolved disks, gradients, and Fourier modes produce their explicit $R_{\text{fixed}, \Theta} \tau$ on the authorized row set; a constant-input pass does not predict preservation of these modes, and procedure-response families are not inferred from the fixed matrix. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-008 – Variable coverage. Fixed-coefficient constant preservation can coexist with spatially varying normalization, support-row membership, covariance, and nonconstant-template response; a row that fails support is absent rather than measured zero. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-009 – Finite boundary. At an admitted finite edge, no contribution wraps or clamps; local normalization may preserve a constant while compact-source peak or integrated response changes. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-010 – Coefficient states. The admitted uniform family yields exact dimensionless $\gamma_i = 1$ only for an explicit selection with complete PTC publication, exact target/payload/occurrence association, MAP permission, and a requested, applicable, eligible, realized named handoff under the exact compatible source binding. Registry presence alone realizes nothing. Missing selection or handoff remains unavailable with no fallback. Structural availability is independent of numerical class; MAP alone classifies finite positive, exact zero, negative finite, non-finite, and unrepresentable. Zero is noncontributing and the other nonpositive/non-finite classes are invalid with exact cause; uniform one is not precision. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-011 – Use-qualified exclusion and non-finite states. A candidate whose typed gate record projects to Boolean nonmembership does not contaminate an accumulator. Structural exclusion prevents payload retrieval; structurally admitted payloads are read only for safe MAP numerical classification before membership and arithmetic. Its complete named-field or other typed record, immutable reason, and cause remain preserved; no universal producer-flag action is inferred. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-012 – Threshold selector. For each predeclared positive finite coefficient population, the selected zero-based index is exactly $\lfloor ([0.75N] + N)/2 \rfloor$; empty input yields $Q_{\star} = 0$, but no row gains support without its own finite positive Q_p . At exact $c = 0$, both relative thresholds are zero and the finite-positive requirement still excludes zero, negative, non-finite, or absent normalization. Values $0 < c \leq 1$ use the same inclusive comparisons in the ordinary recommended range. An explicitly requested expert $c > 1$ may yield empty science support. Negative, non-finite, missing, or unrepresentable c fails before support rows or required-product mutation, with all four exact state/value stages retained. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-013 – Missing response. Every valid map discloses its known MAP-local realized operator response on the exact signal rows. Missing upstream response yields a separately typed limited or unavailable upstream-conditioned state on the complete signal membership, never a smaller hidden subset, zero, or identity. Fixed-state, PTC full-procedure, and PTC+MAP re-resolved families remain distinct; whole-chain RTC-to-CAL-to-PTC-to-MAP response remains unavailable unless separately established. The base MAP bundle remains usable for claims that do not require the absent response; any later response estimate or corrected map has a new versioned identity bound to the original. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-014 – Atomic rejection. A missing required companion or any incompatible identity, unit, response, WCS, shape, policy, or parent rejects the entire observation before every coadd accumulator and count, including apparently compatible pixels, is changed. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-015 – One-observation coadd. A coadd containing one admitted observation under the uniform dimensionless coefficient family reproduces its signal on corresponding centered-integer authorized rows. Conforming in-memory and persisted logical bundles give the same admission, identity, and arithmetic. It reproduces upstream-conditioned response only when the exact response companion is available; otherwise that response remains typed unavailable while the MAP-local operator disclosure remains known. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-016 – Uniform observation coefficients. On each authorized coadd row, two compatible observations produce their equal-observation arithmetic mean and count-like normalization under `SCI-MAP:uniform_observation_coadd_coefficient@1`. The normalization is not precision. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-017 – Missing or invalid observation. An unsupported observation row contributes at no coadd row; an incompletely admitted observation contributes nowhere and does not increment admitted-observation count. Neither case creates a zero-valued scientific substitute. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-018 – Centered integer placement. A canonical target grid requested before construction yields observation maps already bound to that grid. Other compatible shape differences that are nonnegative and even produce the exact half-difference row/column offset; odd, negative, fractional, or WCS-inconsistent relations fail with no mutation. A later crop is a new row-selected product with its own identity and lineage and cannot relabel its parent as compatible. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-019 – Cross-observation covariance. For two maps with variances σ_1^2, σ_2^2 and covariance c , a complete covariance-qualified uniform coadd includes c . If a required block is missing, covariance is partial, symbolic, summarized, lineage-resolvable, or unavailable rather than a complete numerical matrix; the missing block is never zero or independence, and equal-observation normalization is not precision. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-020 – Fixed noise operator. A deterministic admitted realization propagated with fixed state yields the same support-row identities and coefficient-by-coefficient $A_{\text{MAP}, \Pi} \equiv A_{\text{out}}$ and B_{out} as the signal path; recomputed state exposes a named difference. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-021 – Sequential and parallel reduction. Integer facts, identities, masks, support, and cardinality agree exactly. Floating sums agree with a preregistered operation-count/conditioning bound that is frozen before results are viewed and is not loosened after failure. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-022 – No false significance. A deliberate attempt to label signal times square root of merely formal weight as empirical significance is rejected or remains unmistakably labeled formal standardized signal. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-023 – No validity promotion. A downstream product may have its own use-local validity or eligibility but cannot change MAP-local base-product validity or sever the base/unfiltered MAP parent. *Primary*

claim layer: Engineering conformance.

SCI-MAP-PRED-024 – No ordinary predicate for JINC. A deliberate attempt to apply positive-coefficient ordinary admission, support, or complete-bundle availability to JINC is rejected as outside SCI-MAP v0.2. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-025 – Aggregate and index failure. An exact-zero, non-finite, overflowed, or unrepresentable aggregate/index never becomes a live valid product; failure occurs at the declared scope before mutation or completion. *Primary claim layer: Engineering conformance.*

E Product vocabulary

Product	Unit	Contract meaning
Signal numerator	coefficient times mJy/beam	Pre-normalized accumulator; not a sky estimator.
Normalization	declared gridding unit	Positive denominator; precision only under Requirements 020–021.
Normalized signal	mJy/beam	Conditional ordinary map estimand whose vector contains exactly the effective-policy-authorized output rows; no unsupported zero-filled semantic extension.
Formal diagonal weight	inverse square signal unit	Conditional marginal inverse variance under named assumptions; not empirical significance.
Conditional covariance	square signal unit	Full fixed-state noise propagation on the exact scientific output-row domain, with omissions and representation status named.
Response/kernel	mJy/beam when unit-bearing	Realized template response only when the sample-parent identity is proven; otherwise tracer or unavailable.
Geometric/estimator hits	integer	Counts of declared populations, not coverage, precision, or validity.
<code>upstream_eligible_</code> <code>original_footprint_</code> <code>exposure</code>	named time/accounting unit	Unique-original acquisition-footprint accounting for route-candidate ancestors; explicit prose alias: upstream-eligible exposure; not precision.
<code>retained_original_</code> <code>footprint_exposure</code>	named time/accounting unit	Unique-original acquisition-footprint accounting for estimator-admitted ancestors; explicit prose alias: retained exposure; not precision.
Support facts	logical	Separate normalization and science-policy predicates; their effective conjunction defines output-row membership.
MAP-local base-product validity	logical	Authority for the base/unfiltered MAP bundle only; downstream uses own their eligibility.

F Admitted source register and independence statement

The scientific author used only the owner-approved, content-bound packet:

Source	Exact admitted identity
Scope Brief	Approved exact 2026-09-07 brief; SHA-256 0b0504ccdcda3d5844c85f7cc563fb81b47f9bb9fb28dda6438cbc8bd368a8bc.
Author manifest	56 exact references; SHA-256 83baf583f73784bebc6724025fd110e5545303f303c70af590b53c5696ee2c4f; every listed file verified.
MAP successor decisions	2026-08-28 approved post-freeze record; SHA-256 4d2b857b7ec9efe489fe065f464df4ecd23b57a4c1320cda6a10a56592825d1c.
Frozen MAP predecessor	SCI-MAP v0.1/r0.7.1 freeze, source manifest, shared wrapper and six common modules; exact hashes are in the approved 56-reference manifest.
Uniform coefficient	SCI-PTC:analysis_gridding_coefficients@draft-0.1, SCI-PTC:uniform_constant@draft-0.1, and SCI-PTC:uniform_coefficient_handoff@draft-0.1; approved common-source SHA-256 8bc42deb69e23c757883834f7cdb82b94e89210015333f483d15c32377a56685.
PTC-to-MAP boundary	SCI-PTC_TO_SCI-MAP_BOUNDARY v0.2-draft.1/r0.4; SHA-256 58291e1c879dc9116e81a9e3fd4323caf79a176c51e296d8119efb6937272ef8.
MAP/AST/VAL profiles	SCI-MAP:map_upstream_admission@2, SCI-MAP:one_hot_containing_pixel@1, SCI-MAP:observation_coadd_admission@1, SCI-MAP:uniform_observation_coadd_coefficient@1, SCI-AST_TO_SCI-MAP_ORIGINAL_FOOTPRINT_COORDINATE v0.1/r0.1, and the exact admitted uniform VAL Registry/source-binding generation.
Adjacent limits	Exact approved PTC, AST, VAL, and NOI freezes/requirements enumerated in the 56-reference manifest; no adjacent science is expanded.

No implementation, test, audit, repair, validation, reduction, production- status record, manager dossier, unlisted link, repository history, sibling repository, or web scientific source entered this authorship. The frozen ordinary derivation and admitted uniform family were incorporated rather than rederived. The future canonical commit and final VAL/source binding remain pending and are not invented here.